

Advanced AC Circuit

Frequency Response

Words in Control System

Household appliances:

- heating and air-conditioning systems,
- switch-controlled thermostats,
- washers and dryers,
- cruise controllers in automobiles,
- elevators,
- traffic lights,
- manufacturing plants,
- navigation systems

In the aerospace field

In the manufacturing sector...

—Combination of circuit theory and communication theory

Outline

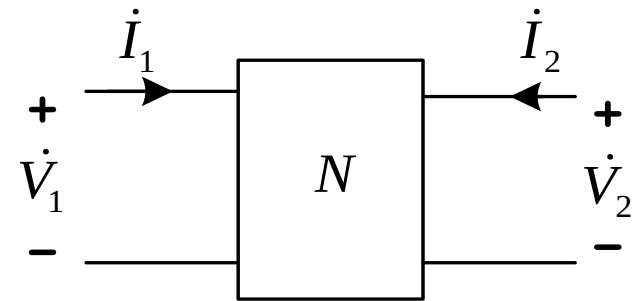
- **Transfer Function and Frequency Response**
- **Filters**
- **Series/Parallel Resonance**

Transfer Function (Network Function)

- **Transfer function**

— the frequency-dependent ratio of a phasor output $Y(\omega)$ to a phasor input $X(\omega)$.

$$H(\omega) = \frac{\dot{Y}(\omega)}{\dot{X}(\omega)}$$



- **Driving Point Function**

— the output and input are in the same port.

$$Z(\omega) = \frac{\dot{V}_1}{\dot{I}_1}$$

— Driving point impedance

$$H(\omega) = \frac{\dot{I}_1}{\dot{V}_1}$$

— Driving point admittance

Transfer Function (Network Function)

- Transfer Function**

— the output and input are in the different ports.

$$H(\omega) = \frac{\dot{V}_2}{\dot{V}_1}$$

— Voltage Gain

$$H(\omega) = \frac{\dot{I}_2}{\dot{I}_1}$$

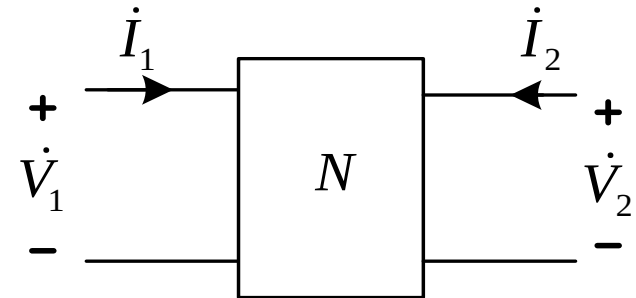
— Current Gain

$$H(\omega) = \frac{\dot{V}_2}{\dot{I}_1}$$

— Transfer impedance

$$H(\omega) = \frac{\dot{I}_2}{\dot{V}_1}$$

— Transfer admittance



Transfer Function (Network Function)

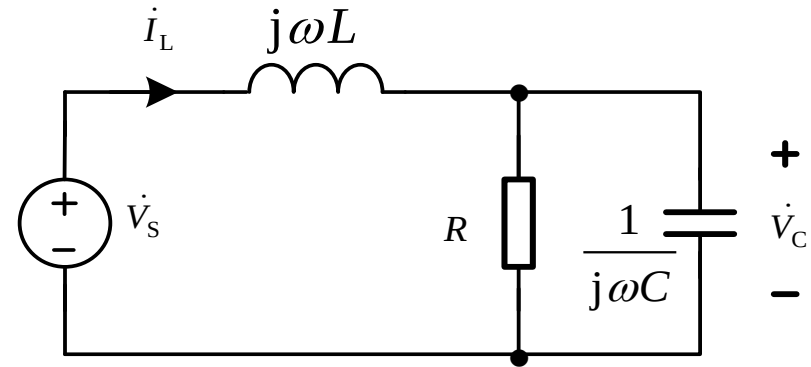
Example

Obtain the transfer function V_C/V_S

Solution

$$H(\omega) = \frac{\dot{V}_C}{\dot{V}_S}$$

$$H(\omega) = \frac{\dot{V}_C}{\dot{V}_S} = \frac{\frac{R / j\omega C}{R + 1 / j\omega C}}{j\omega L + \frac{R / j\omega C}{R + 1 / j\omega C}} = \frac{R}{LRC(j\omega)^2 + Lj\omega + R} = \frac{1 / LC}{(j\omega)^2 + j\omega / RC + 1 / LC}$$



Transfer Function is a ratio of two polynomials of $j\omega$

$H(\omega)$ can also be written as $H(j\omega)$.

Frequency Response

The frequency response of a circuit

—the variation in its behavior with change in signal frequency.

- **Magnitude Response**

$$a(\omega) = |H(j\omega)| = Y_m(\omega) / X_m(\omega)$$

- **Phase Response**

$$\theta(\omega) = \angle H(j\omega) = \phi_y - \phi_x$$

- **Magnitude Response Plot:** $a(\omega) \sim \omega$
- **Phase Response Plot:** $\theta(\omega) \sim \omega$

Frequency Response

Example

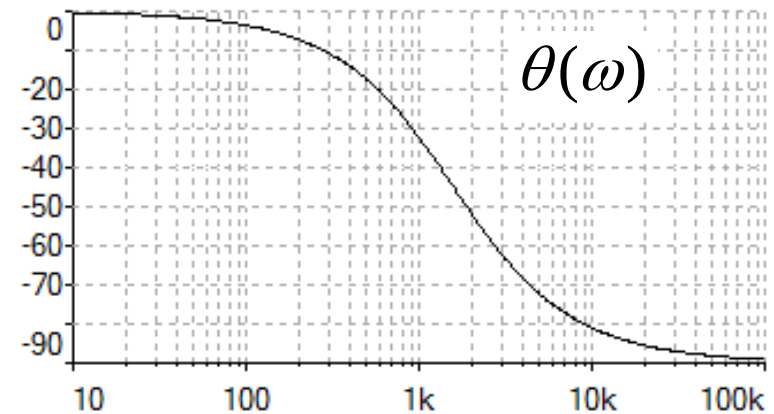
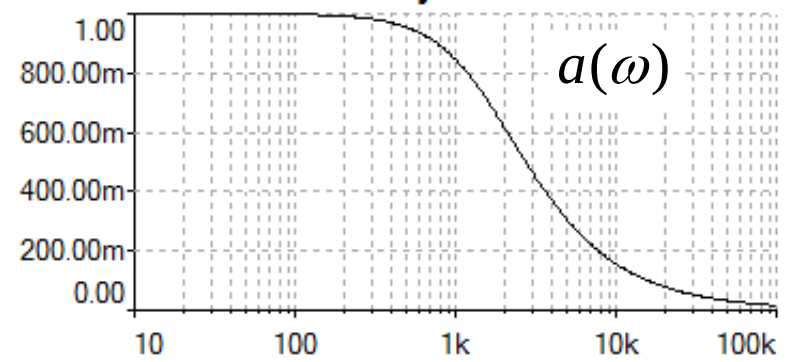
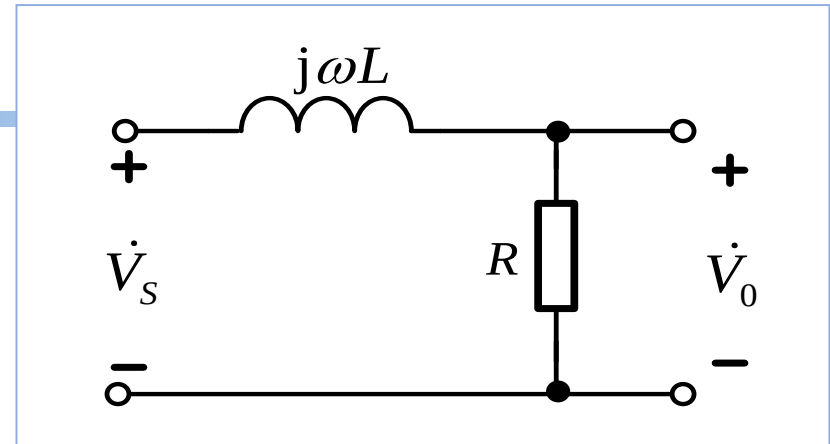
$$R = 100\Omega \quad L = 0.01\text{H}$$

$$H(j\omega) = \frac{\dot{V}_o}{\dot{V}_s} = \frac{R}{j\omega L + R} = \frac{R/L}{j\omega + R/L}$$

$$H(j\omega) = \frac{10^4}{j\omega + 10^4}$$

$$a(\omega) = |H(j\omega)| = \frac{1}{\sqrt{1 + \omega^2 / 10^8}}$$

$$\theta(\omega) = \angle H(j\omega) = -\tan^{-1} \frac{\omega}{10^4}$$



The Decibel Scale

- **Bel — measurement for Power Gain**

$$G_{\text{bel}} = \log_{10} \frac{P_2}{P_1}$$

- **Decibel — 1/10th of a bel**

$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1}$$

$$\text{When } P_2 = P_1 \quad G_{\text{dB}} = 10 \lg \frac{P_2}{P_1} = 10 \lg 1 = 0 \text{ dB}$$

$$\text{When } P_2 = 2P_1 \quad G_{\text{dB}} = 10 \lg \frac{P_2}{P_1} = 10 \lg 2 = 3 \text{ dB}$$

$$\text{When } P_2 = 0.5P_1 \quad G_{\text{dB}} = 10 \lg \frac{P_2}{P_1} = 10 \lg 0.5 = -3 \text{ dB}$$

The Decibel Scale

- Decibel — comparing for levels of voltage or current (on one resistor)**

$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1} = 10 \lg \frac{V_{2\text{m}}^2 / 2R}{V_{1\text{m}}^2 / 2R} = 20 \lg \frac{V_{2\text{m}}}{V_{1\text{m}}}$$

$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1} = 10 \lg \frac{I_{2\text{m}}^2 R / 2}{I_{1\text{m}}^2 R / 2} = 20 \lg \frac{I_{2\text{m}}}{I_{1\text{m}}}$$

When $P_2 = P_1$ $G_{\text{dB}} = 10 \lg \frac{P_2}{P_1} = 10 \lg 1 = 0 \text{ dB}$ $V_{2\text{m}} = V_{1\text{m}}$

When $P_2 = 2P_1$ $G_{\text{dB}} = 10 \lg \frac{P_2}{P_1} = 10 \lg 2 = 3 \text{ dB}$ $V_{2\text{m}} = \sqrt{2} V_{1\text{m}}$

When $P_2 = 0.5P_1$ $G_{\text{dB}} = 10 \lg \frac{P_2}{P_1} = 10 \lg 0.5 = -3 \text{ dB}$ $V_{2\text{m}} = V_{1\text{m}} / \sqrt{2}$

Code Plots

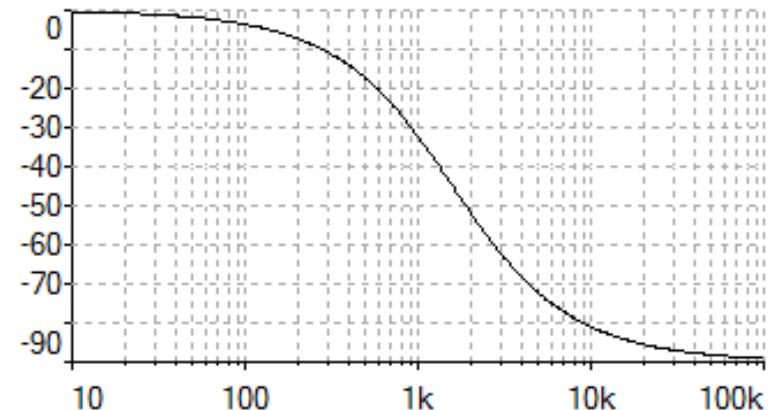
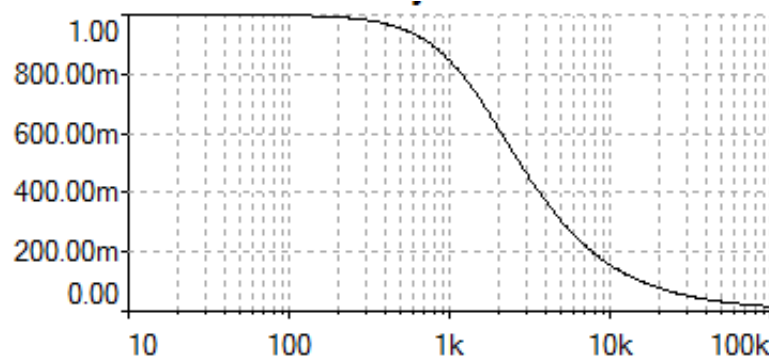
Code Plots

—Semi-logarithm plots of the magnitude (in ratio value) and phase (in degrees) of a transfer function versus frequency.

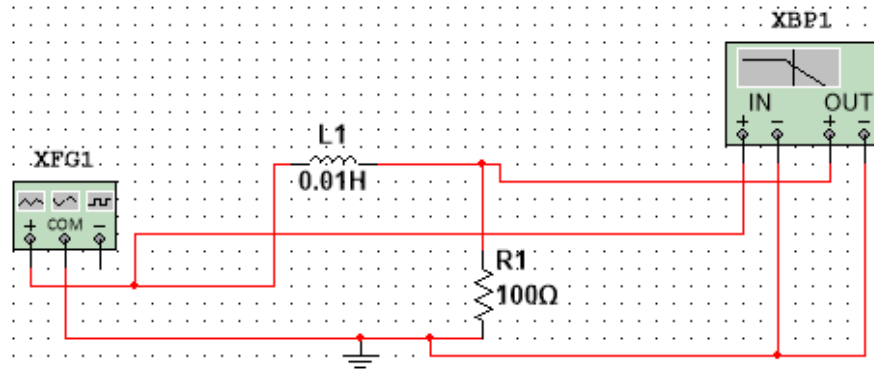
Magnitude axis — linear scale

Phase axis — linear scale

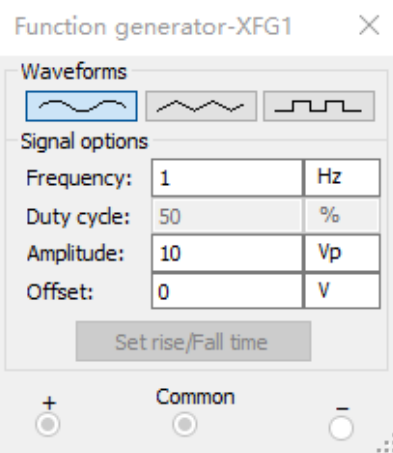
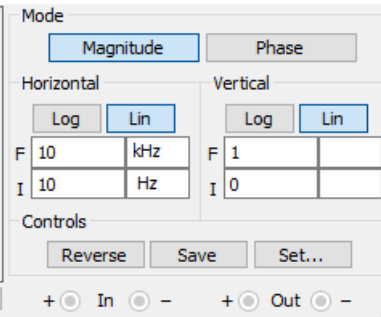
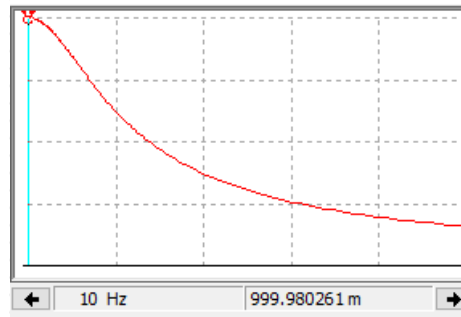
Frequency axis — logarithmic scale



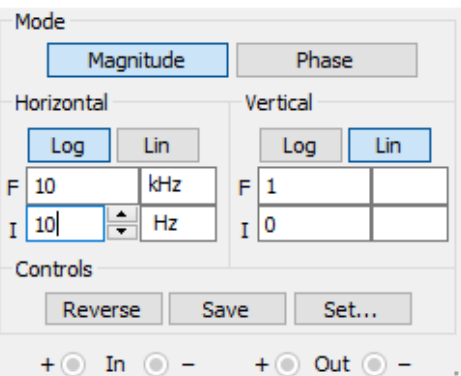
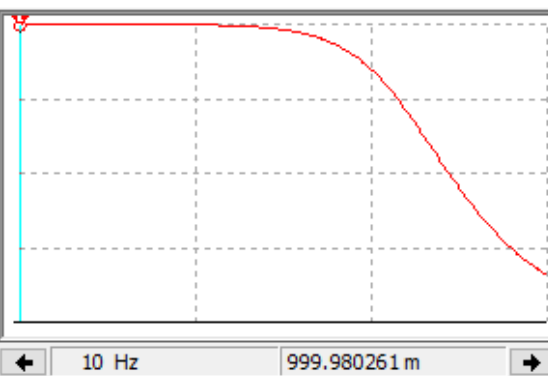
Code Plot Simulation



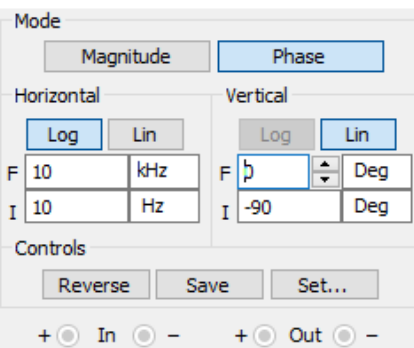
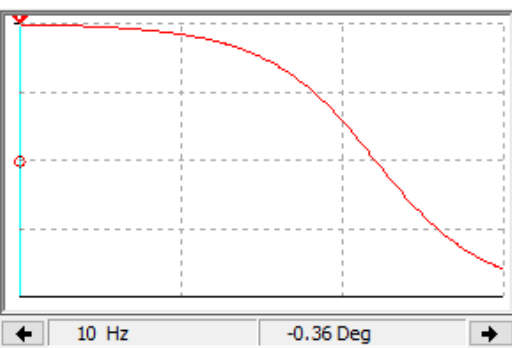
Bode Plotter-XBP1



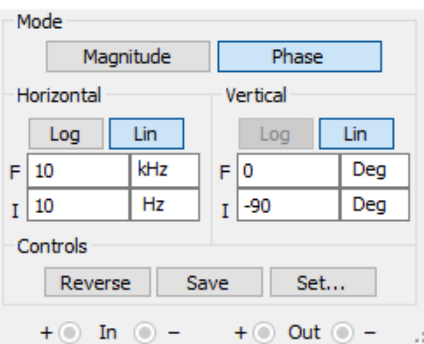
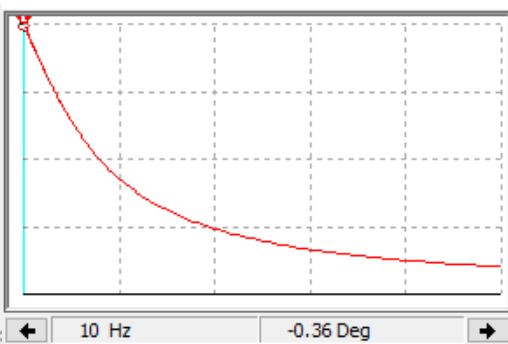
Bode Plotter-XBP1



Bode Plotter-XBP1



Bode Plotter-XBP1



Homework

Problem 1, 2, 3

-
- **Transfer Function and Frequency Response**
 - **Filters**
 - **Series/Parallel Resonance**

Filters

- **Filter**

- a circuit that is designed to pass signals with desired frequencies and reject or attenuate others.

- **Passive Filter**

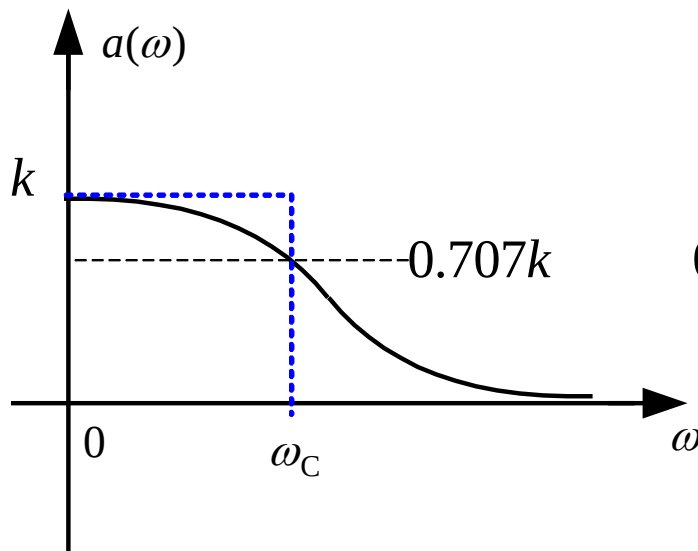
- a filter consists of only passive elements R , L and C .

- **Active Filter**

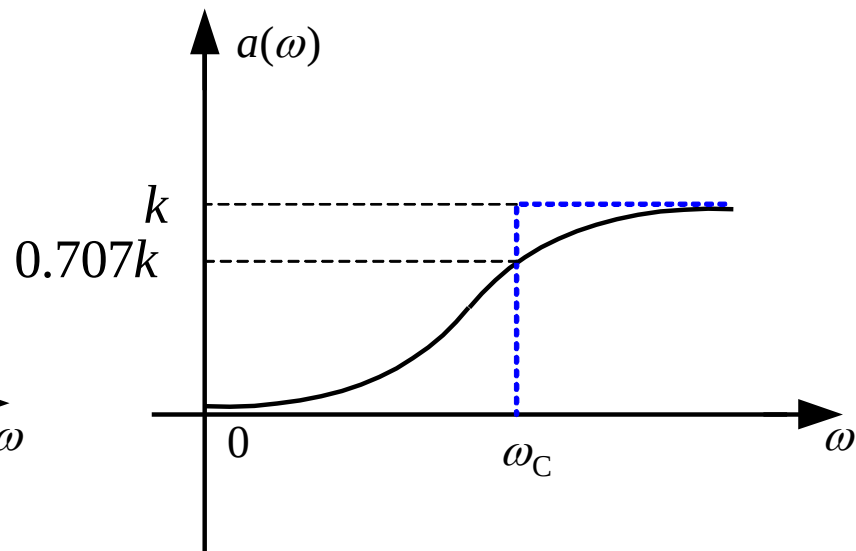
- a filter consists of active elements such as transistors and op amps in addition to passive elements R , L and C .

Types of Filters

Four types of filters according to magnitude response:



Lowpass Filter



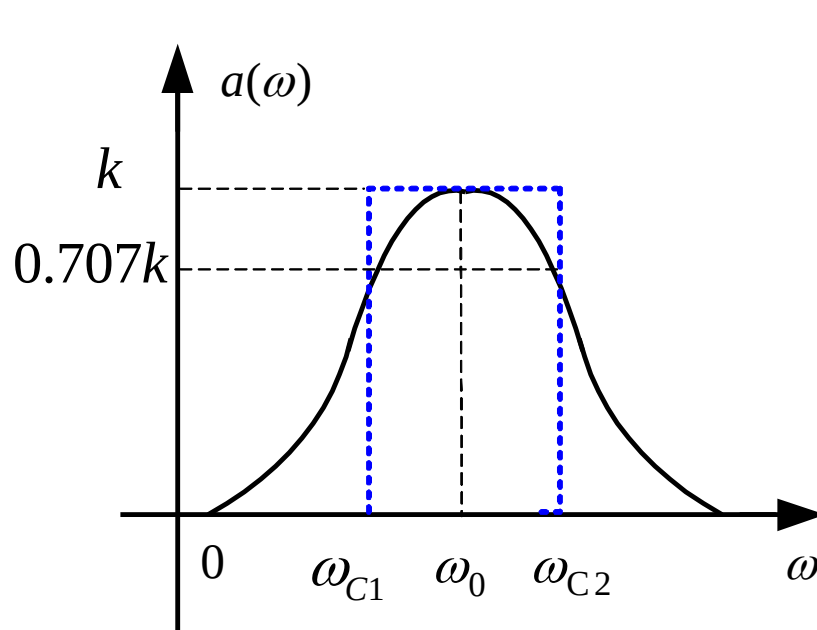
Highpass Filter

- **Cutoff frequency $f_C(\omega_C)$:** the frequency at which the transfer function H drops in magnitude to 70.71% of its maximum value.

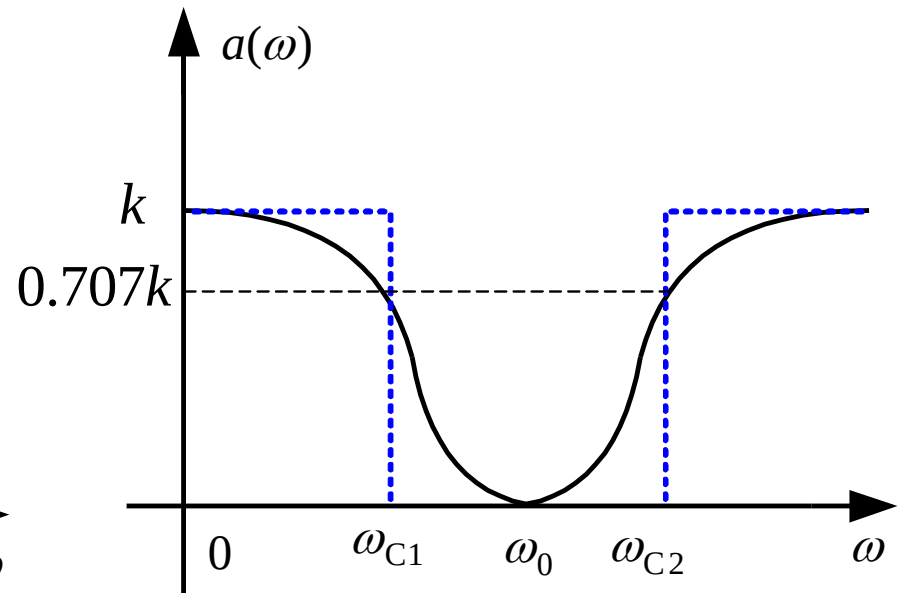
Types of Filters

- Center frequency $f_0(\omega_0)$

- Rejection frequency $f_0(\omega_0)$



Bandpass Filter

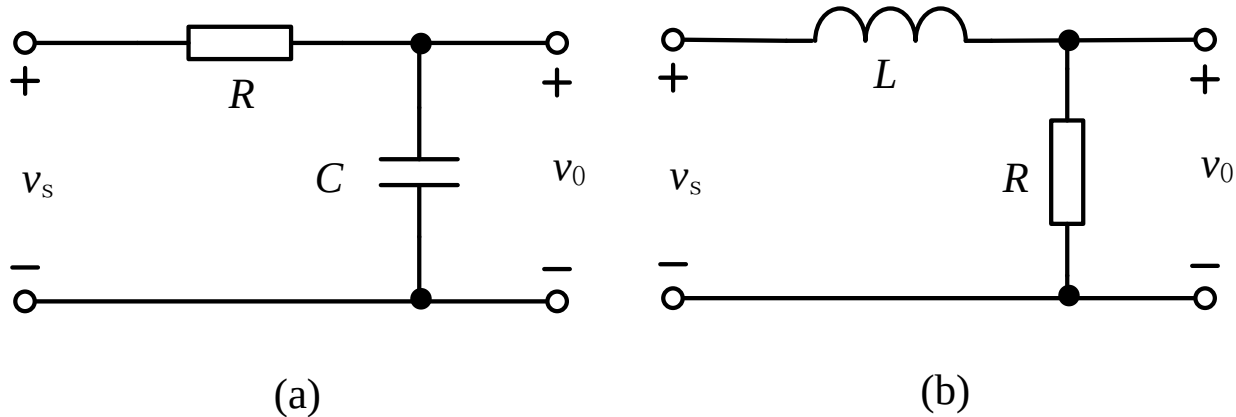


Bandstop Filter

k : Gain in band

The First Order Lowpass

- Example



$$(a) \quad H(j\omega) = \frac{\dot{V}_o}{\dot{V}_s} = \frac{1/j\omega C}{1/j\omega C + R} = \frac{1/RC}{j\omega + 1/CR} = \frac{1}{1 + j\omega CR} \quad \omega_c = \frac{1}{RC}$$

$$(b) \quad H(j\omega) = \frac{\dot{V}_o}{\dot{V}_s} = \frac{R}{j\omega L + R} = \frac{R/L}{j\omega + R/L} = \frac{1}{1 + j\omega L/R} \quad \omega_c = \frac{R}{L}$$

$$k = 1$$

$$H(j\omega) = \frac{1}{1 + j\omega / \omega_c}$$

The First Order Lowpass

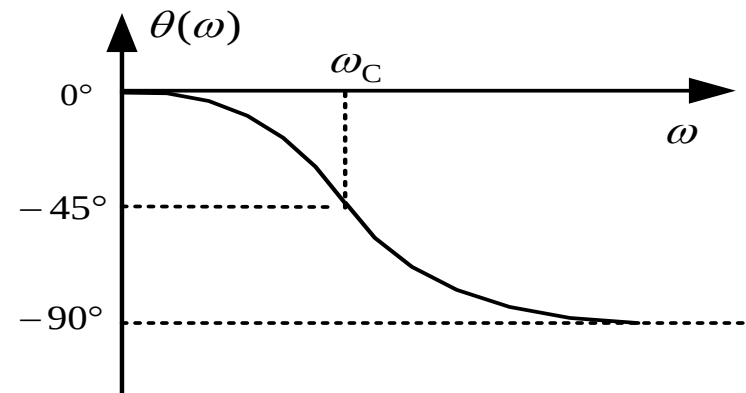
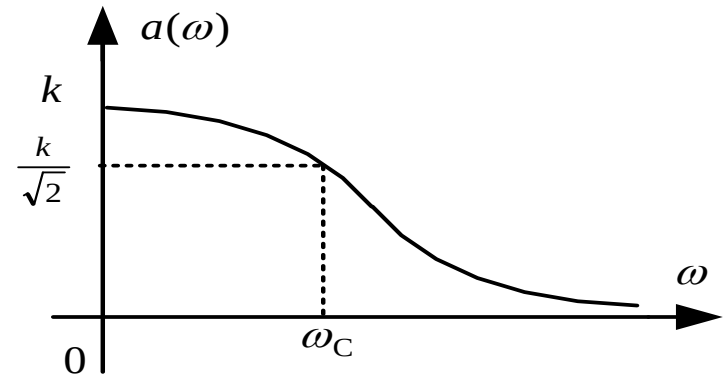
- General network Function for 1st order lowpass filter

$$H(j\omega) = \frac{k\omega_c}{j\omega + \omega_c} = \frac{k}{1 + j\frac{\omega}{\omega_c}}$$

- Frequency response

$$a(\omega) = \frac{|k|}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^2}}$$

$$\theta(\omega) = -\tan^{-1} \frac{\omega}{\omega_c}$$



The First Order Lowpass

- Example:** Demonstrate the circuit can be used as a lowpass filter. Select proper element value so that the band gain is 4 and cutoff angular frequency is 100rad/s.

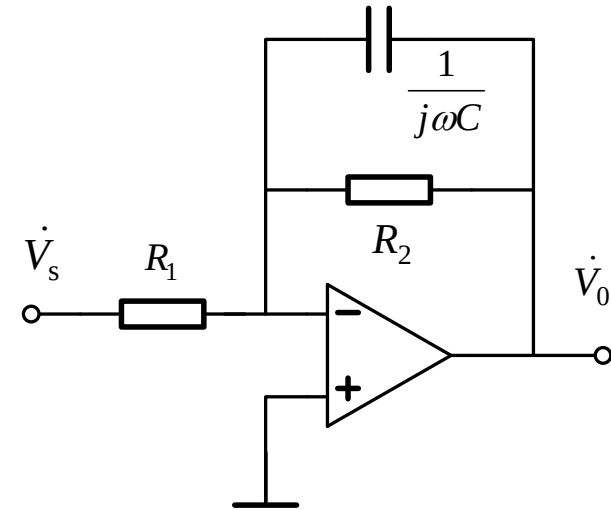
Solution Applying KCL at inverse input terminal, get

$$\frac{\dot{V}_s}{R_1} = -\dot{V}_o \left(\frac{1}{R_2} + j\omega C \right)$$

$$H(\omega) = \frac{\dot{V}_o}{\dot{V}_s} = -\frac{R_2}{R_1} \cdot \frac{1}{1 + j\omega CR_2} = k \cdot \frac{1}{1 + j\frac{\omega}{\omega_c}}$$

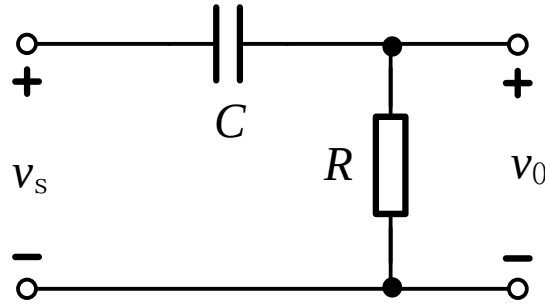
$$k = -\frac{R_2}{R_1} \quad \omega_c = \frac{1}{R_2 C} \quad |k| = \frac{R_2}{R_1} = 4$$

Select $R_1 = 10\text{k}\Omega$ $R_2 = 40\text{k}\Omega$ $C = \frac{1}{R_2 \omega_c} = 250\text{nF}$

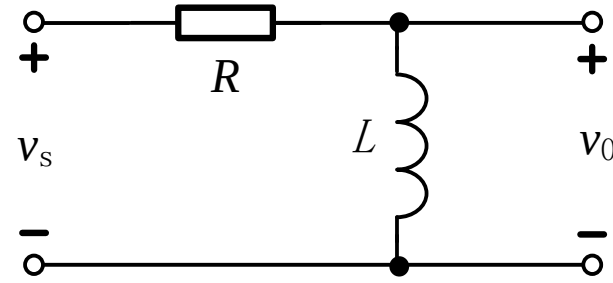


The First Order Highpass

- Example



(a)



(b)

$$(a) \quad H(\omega) = \frac{R}{R + \frac{1}{j\omega C}} = \frac{j\omega}{j\omega + 1/RC} = \frac{j\omega CR}{1 + j\omega CR} \quad \omega_c = \frac{1}{RC}$$

$$(b) \quad H(\omega) = \frac{j\omega L}{R + j\omega L} = \frac{j\omega}{j\omega + R/L} = \frac{j\omega L/R}{1 + j\omega L/R} \quad \omega_c = \frac{R}{L}$$

$$k = 1$$

$$H(j\omega) = \frac{j\frac{\omega}{\omega_c}}{1 + j\frac{\omega}{\omega_c}}$$

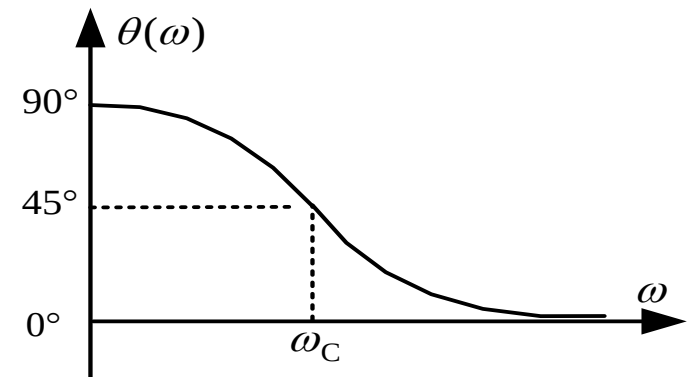
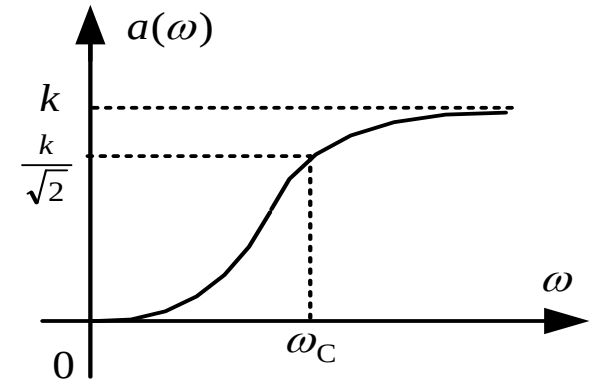
The First Order Highpass

- General network Function for 1st order highpass filter

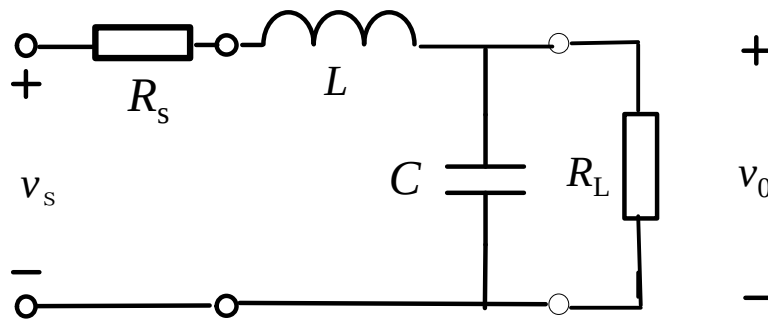
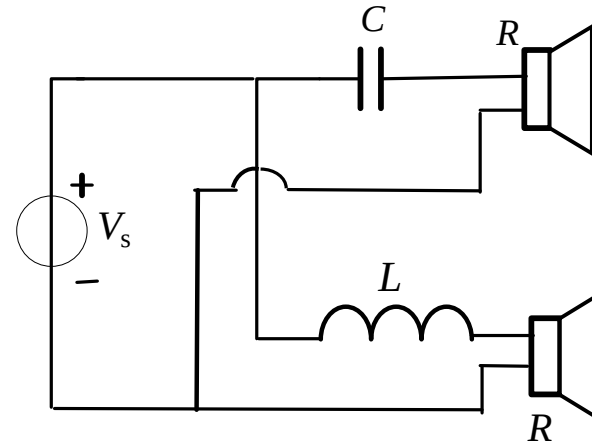
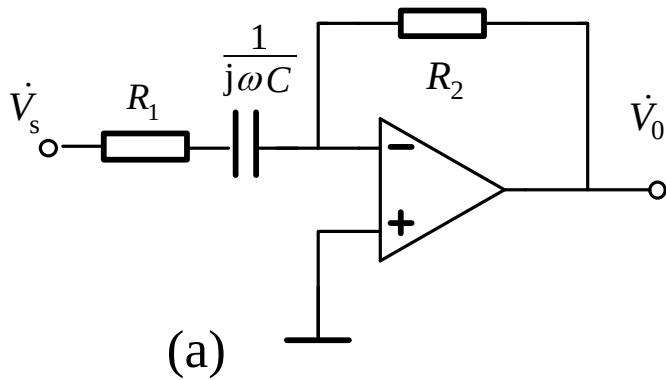
$$H(j\omega) = k \frac{j\omega}{j\omega + \omega_c} = k \frac{j\frac{\omega}{\omega_c}}{1 + j\frac{\omega}{\omega_c}}$$

$$a(\omega) = \frac{k}{\sqrt{1 + \left(\frac{\omega_c}{\omega}\right)^2}}$$

$$\theta(\omega) = \tan^{-1} \frac{\omega_c}{\omega}$$



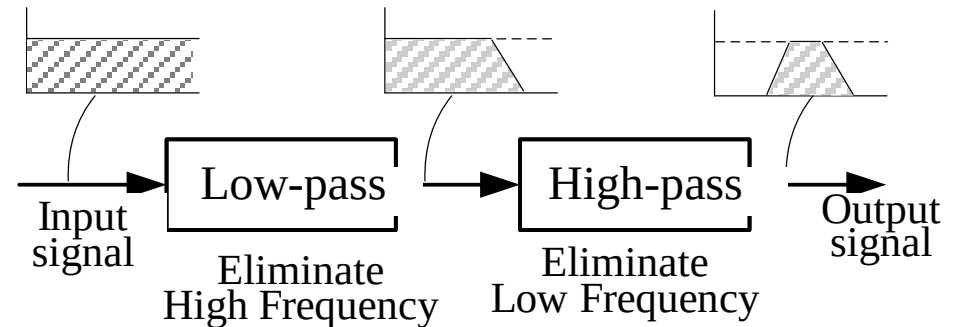
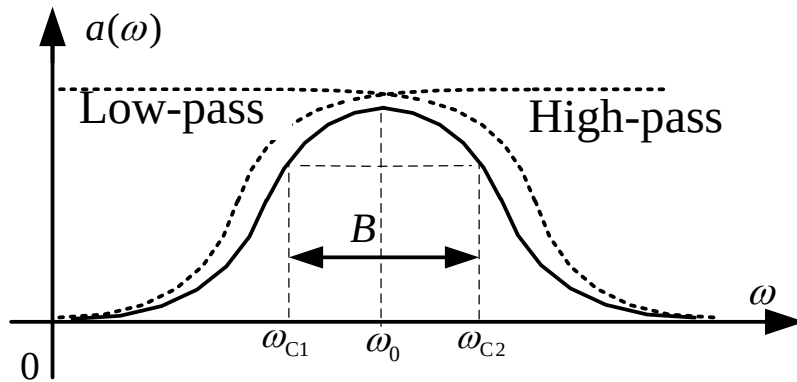
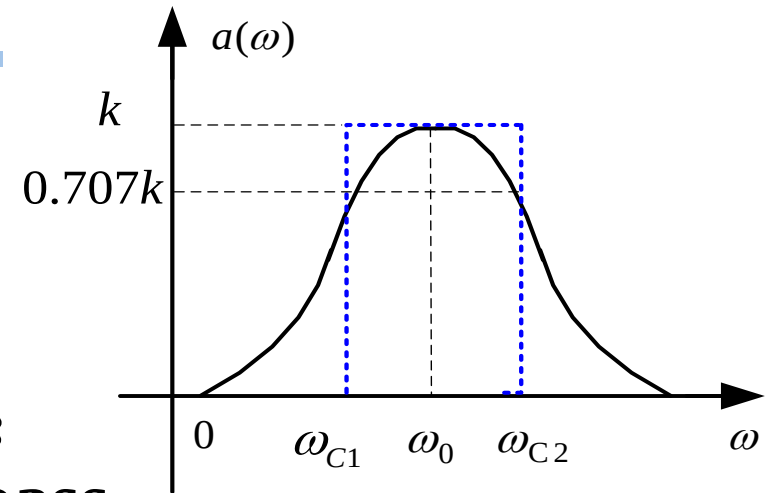
Exercises: Analyze the frequency response.



$$a(\omega) = \frac{k}{\sqrt{1 + (\omega / \omega_c)^{2N}}}$$

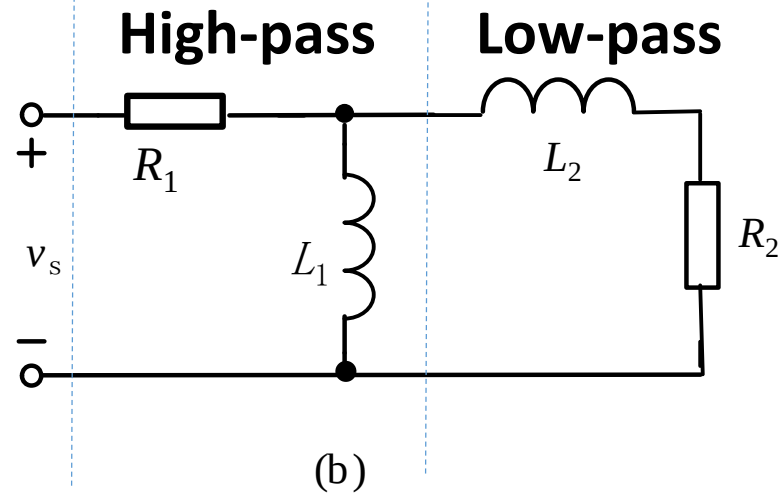
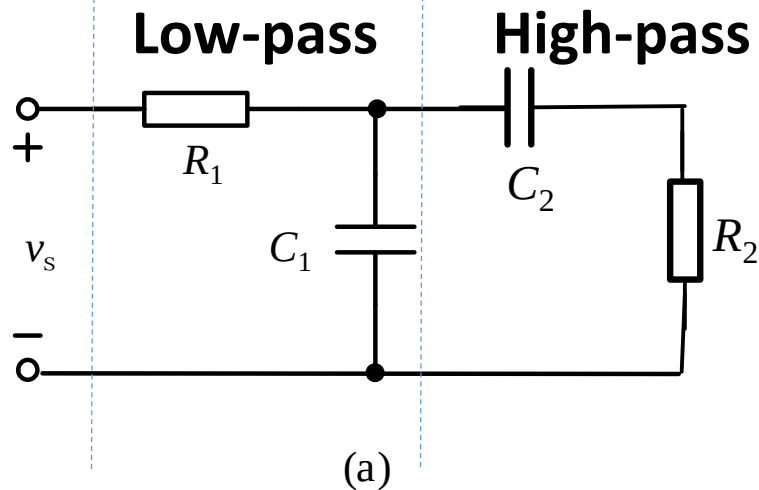
The 2nd Order Band-pass

- A band-pass filter is designed to pass all frequencies within a band of $\omega_{c1} < \omega < \omega_{c2}$
- A method to obtain a band-pass: cascading a low-pass and a high-pass



- **NOTE:** $f_{c \text{ low-pass}} > f_{c \text{ high-pass}}$

The 2nd Order Band-pass



$$(a) \quad \dot{V}_o = \dot{V}_2 = \frac{Z_{C_1} // (Z_{C_2} + R_2)}{R_1 + Z_{C_1} // (Z_{C_2} + R_2)} \frac{R_2}{(Z_{C_2} + R_2)} \dot{V}_s$$

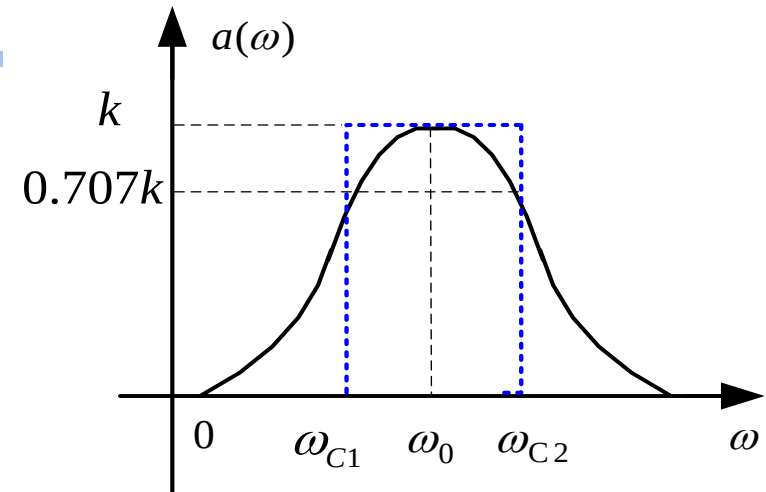
$$H_C = \frac{\dot{V}_2}{\dot{V}_s} = \frac{j\omega / C_1 R_1}{(j\omega)^2 + j\omega \left(\frac{1}{C_1 R_2} + \frac{1}{C_1 C_2} + \frac{1}{C_2 R_2} \right) + \frac{1}{C_1 R_1 C_2 R_2}}$$

$$(b) \quad H_L = \frac{\dot{V}_2}{\dot{V}_s} = \frac{j\omega R_2 / L_2}{(j\omega)^2 + j\omega \left(\frac{R_1}{L_2} + \frac{R_1}{L_1} + \frac{R_2}{L_2} \right) + \frac{R_1 R_2}{L_1 L_2}}$$

The 2nd Order Band-pass

- General network Function for 2nd order band-pass filter

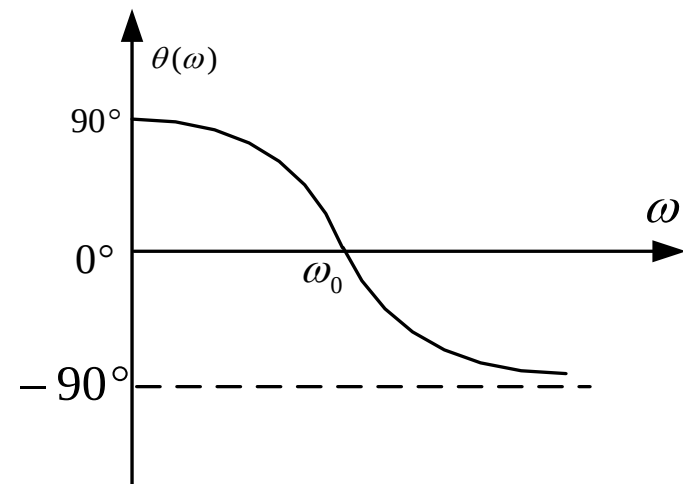
$$H(j\omega) = k \frac{\frac{\omega_0}{Q} j\omega}{(j\omega)^2 + \frac{\omega_0}{Q} j\omega + \omega_0^2}$$



k : Passband Gain ω_0 : Center Frequency, Q : Quality Factor

$$a(\omega) = \frac{k}{\sqrt{1 + Q^2 \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)^2}}$$

$$\theta(\omega) = -\tan^{-1} Q \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)$$



The 2nd Order Bandpass

- **Quality Factor** $Q = \frac{\omega_0}{B}$

- **Bandwidth** $B = \omega_{c2} - \omega_{c1} = \frac{\omega_0}{Q}$

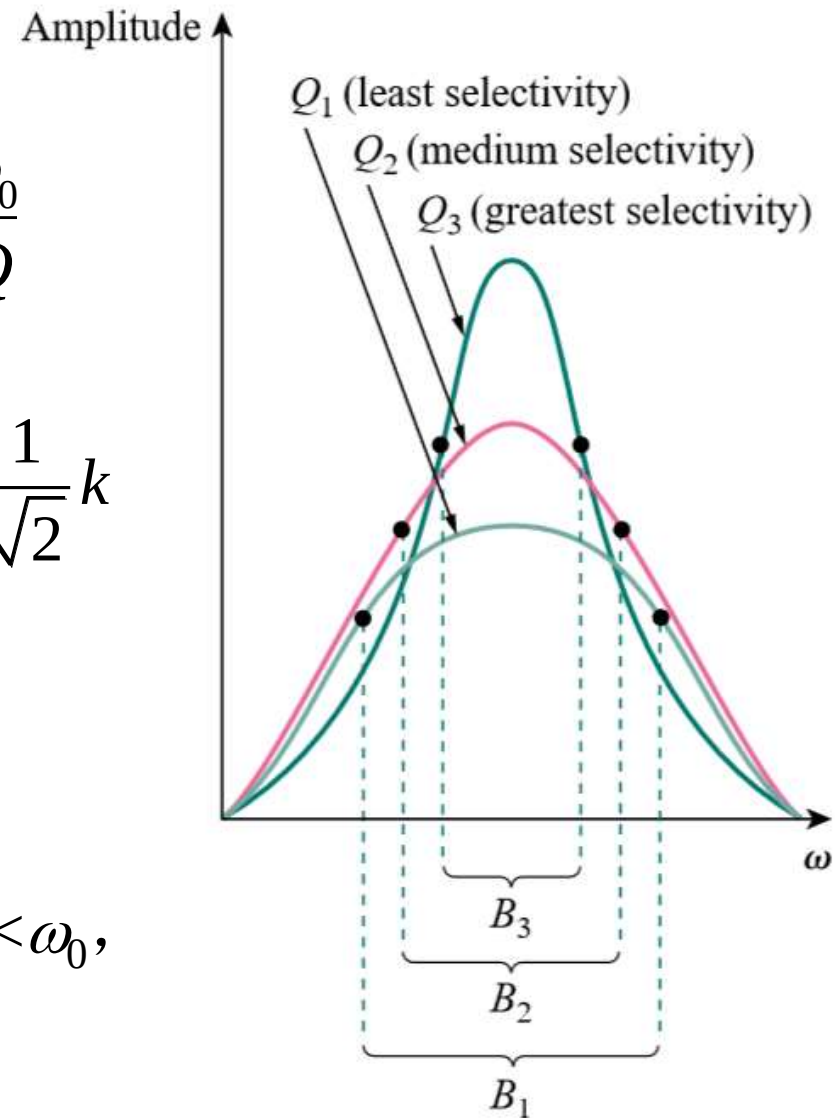
$$\omega_{c1} \cdot \omega_{c2} = \omega_0^2$$

Set $a(\omega_c) = \frac{k}{\sqrt{1 + Q^2 \left(\frac{\omega_c}{\omega_0} - \frac{\omega_0}{\omega_c} \right)^2}} = \frac{1}{\sqrt{2}} k$

get $\omega_{c1}, \omega_{c2} = \frac{\omega_0}{2Q} (\sqrt{1 + 4Q^2} \mp 1)$

For 'high Q ' circuit: $Q > 10$, $B \ll \omega_0$,

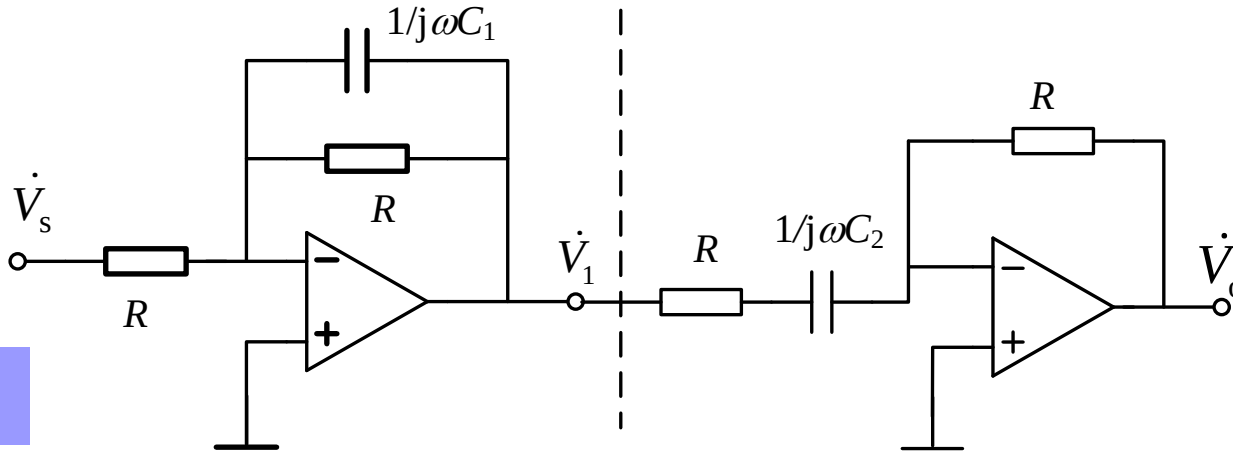
$$\omega_{c1}, \omega_{c2} \approx \omega_0 \mp \frac{B}{2}$$



Example

Given $C_1=100\text{nF}$, $C_2=400\text{nF}$, $R=1\text{k}\Omega$. Find $H(j\omega)$, ω_0 ,

B , Q and ω_{C1} , ω_{C2} .



solution

First stage transfer function

$$H_1(j\omega) = \frac{\dot{V}_1}{\dot{V}_s} = -\frac{1}{1 + j\omega C_1 R} = -\frac{1/RC_1}{j\omega + 1/RC_1}$$

Second stage transfer function

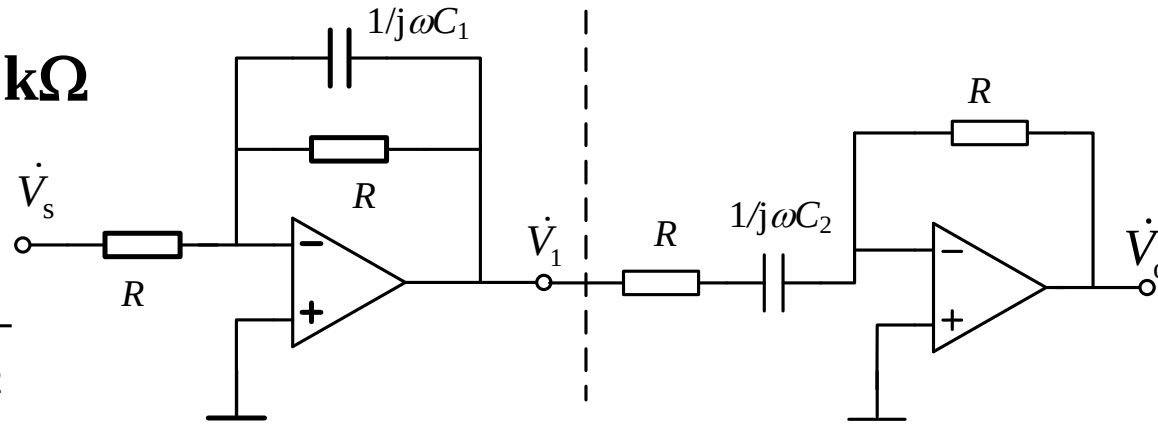
$$H_2(\omega) = \frac{\dot{V}_o}{\dot{V}_1} = -\frac{j\omega C_2 R}{1 + j\omega C_2 R} = -\frac{j\omega}{j\omega + 1/RC_2}$$

Thus

$$H(j\omega) = \frac{\dot{V}_o}{\dot{V}_s} = \frac{\dot{V}_o}{\dot{V}_1} \frac{\dot{V}_1}{\dot{V}_s} = H_1(\omega)H_2(\omega) = \frac{j\omega / RC_1}{(j\omega + 1/RC_1)(j\omega + 1/RC_2)}$$

$$C_1=100\text{nF}, C_2=400\text{nF}, R=1\text{k}\Omega$$

$$H(j\omega) = k \frac{\frac{\omega_0}{Q} j\omega}{(j\omega)^2 + \frac{\omega_0}{Q} j\omega + \omega_0^2}$$



$$H(j\omega) = \frac{j\omega \cdot 10^4}{(j\omega + 10^4)(j\omega + 2500)} = \frac{j\omega \cdot 10^4}{(j\omega)^2 + 12500 \cdot j\omega + 25 \times 10^6}$$

$$\omega_0 = 5000 \text{ rad/s} \left\{ \begin{array}{l} B = \omega_{c2} - \omega_{c1} = \frac{\omega_0}{Q} = 12500 \text{ rad/s} \\ \omega_{c1} \cdot \omega_{c2} = \omega_0^2 \end{array} \right. \quad Q = \frac{\omega_0}{12500} = 0.4$$

$$k = 0.8$$



$$\omega_{c1} = \frac{\omega_0}{2Q} (\sqrt{1 + 4Q^2} - 1) = \frac{\omega_0}{Q} (0.14) = 1750 \text{ rad/s}$$

$$\omega_{c2} = \frac{\omega_0}{2Q} (\sqrt{1 + 4Q^2} + 1) = \frac{\omega_0}{Q} (1.14) = 14250 \text{ rad/s}$$

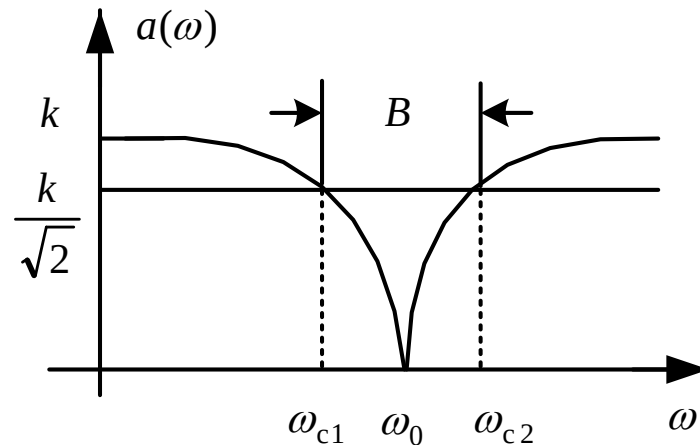
The 2nd Order Bandstop(Bandreject)*

- General network Function for 2nd order bandstop filter

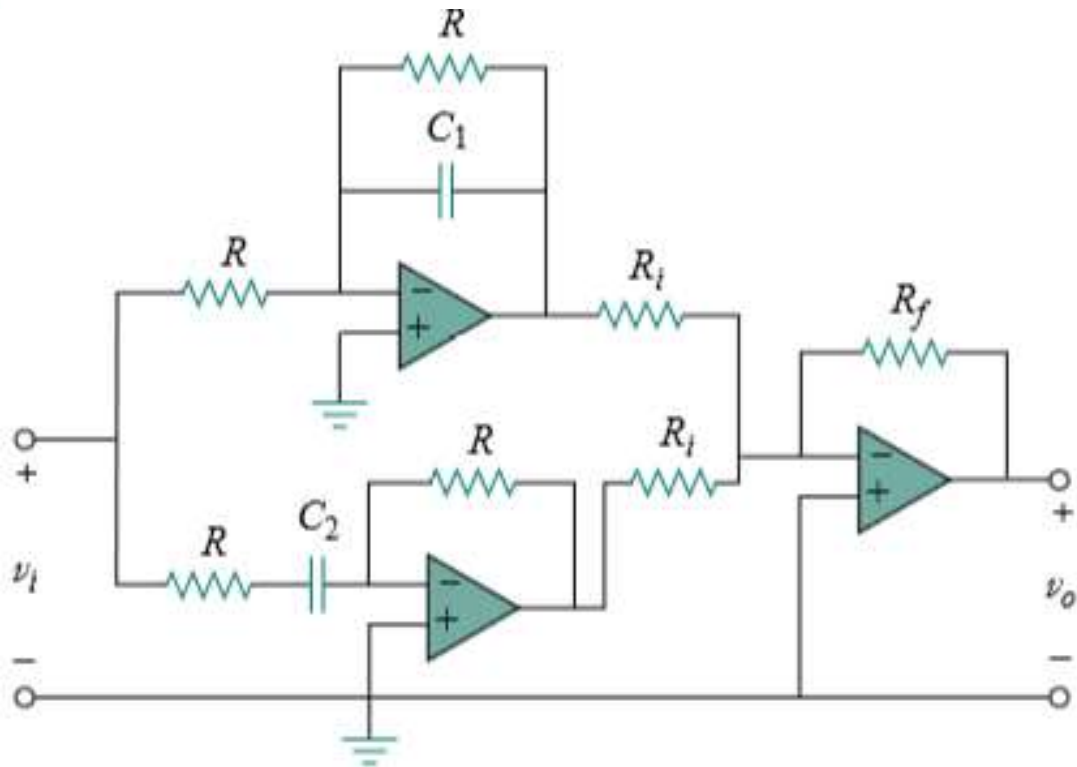
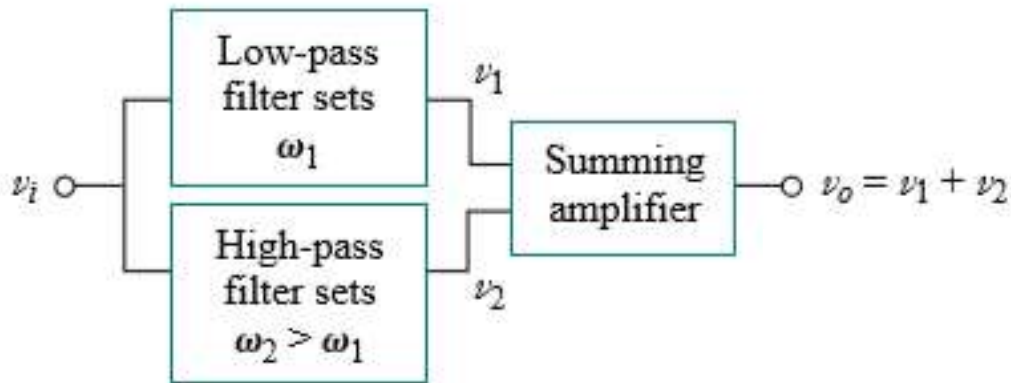
$$H(j\omega) = k \frac{(j\omega)^2 + 2\beta \cdot j\omega + \omega_0^2}{(j\omega)^2 + \frac{\omega_0}{Q} \cdot j\omega + \omega_0^2} \quad \beta \ll \frac{\omega_0}{2Q}$$

- Amplitude frequency response

Stopband width $B = \frac{\omega_0}{Q}$



A Method to Obtain a Band-stop: Paralleling a Low-pass and a High-pass



$$H(j\omega) = \frac{R_f}{R_i} \frac{(j\omega)^2 + 2j\omega\omega_1 + \omega_1\omega_2}{(j\omega)^2 + j\omega(\omega_1 + \omega_2) + \omega_1\omega_2}$$

Homework

- **5, 6, 7, 9, 11,14**

Recap

- Transfer function**

— the frequency-dependent ratio of a phasor output (an element voltage or current) $Y(\omega)$ to a phasor input (source voltage or current) $X(\omega)$.

$$H(\omega) = \frac{\dot{Y}(\omega)}{\dot{X}(\omega)}$$

$$H(\omega) = \text{voltage gain} = \frac{V_o(\omega)}{V_i(\omega)}$$

$$H(\omega) = \text{current gain} = \frac{I_o(\omega)}{I_i(\omega)}$$

$$H(\omega) = \text{transfer impedance} = \frac{V_o(\omega)}{I_i(\omega)}$$

$$H(\omega) = \text{transfer admittance} = \frac{I_o(\omega)}{V_i(\omega)}$$

Recap — Filter

- **1st order low-pass filter**

$$H(j\omega) = \frac{k\omega_c}{j\omega + \omega_c} = \frac{k}{1 + j\frac{\omega}{\omega_c}}$$

- **2nd order low-pass filter**

$$H(j\omega) = \frac{k}{(j\omega)^2 + \frac{\omega_0}{Q}j\omega + \omega_0^2}$$

- **2nd order band-pass filter**

$$H(j\omega) = k \frac{\frac{\omega_0}{Q}j\omega}{(j\omega)^2 + \frac{\omega_0}{Q}j\omega + \omega_0^2}$$

- **1st order high-pass filter**

$$H(j\omega) = k \frac{j\omega}{j\omega + \omega_c} = k \frac{j\frac{\omega}{\omega_c}}{1 + j\frac{\omega}{\omega_c}}$$

- **2nd order high-pass filter**

$$H(j\omega) = \frac{k(j\omega)^2}{(j\omega)^2 + \frac{\omega_0}{Q}j\omega + \omega_0^2}$$

- **2nd order band-stop filter**

$$H(j\omega) = k \frac{(j\omega)^2 + 2\beta \cdot j\omega + \omega_0^2}{(j\omega)^2 + \frac{\omega_0}{Q} \cdot j\omega + \omega_0^2}$$
$$\beta \ll \frac{\omega_0}{2Q}$$

Recap — Filter

Steps to judge the type of a filter:

Find $H(j\omega)$ or $|H(\omega)|$

- $\omega=0, |H(0)| = \alpha(0) = |k|$
 - $\omega \rightarrow \infty, |H(\infty)| = \alpha(\infty) = 0$
- } **Low-pass**

- $\omega=0, |H(0)| = \alpha(0) = 0$
 - $\omega \rightarrow \infty, |H(\infty)| = \alpha(\infty) = |k|$
- } **High-pass**

- $\omega=0, |H(0)| = \alpha(0) = 0$
 - $\omega \rightarrow \infty, |H(\infty)| = \alpha(\infty) = 0$
- } **Band-pass**

- $\omega=0, |H(0)| = \alpha(0) = |k|$
 - $\omega \rightarrow \infty, |H(\infty)| = \alpha(\infty) = |k|$
- } **Band-stop**

Recap — Filter

- **Cutoff frequency $f_c(\omega_c)$:** the frequency at which the transfer function H drops in magnitude to 70.71% of its maximum value.
— **half-power frequencies**

$$|H(\omega_c)| = \frac{1}{\sqrt{2}} |H(0)|$$

Low-pass

$$|H(\omega_c)|_{\text{dB}} - |H(0)|_{\text{dB}} = -3\text{dB}$$

$$|H(\omega_c)| = \frac{1}{\sqrt{2}} |H(\infty)|$$

High-pass

$$|H(\omega_c)|_{\text{dB}} - |H(\infty)|_{\text{dB}} = -3\text{dB}$$

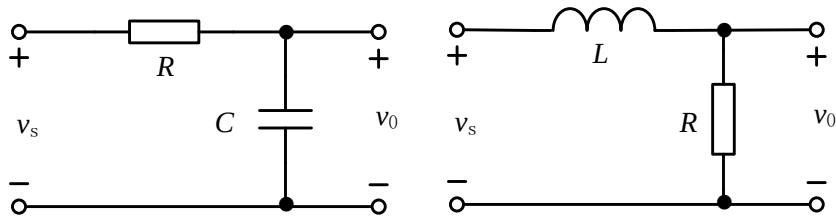
$$|H(\omega_c)| = \frac{1}{\sqrt{2}} |H(\omega_0)|$$

Band-pass

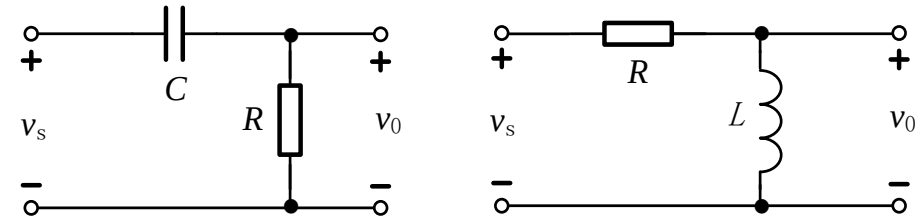
$$|H(\omega_c)|_{\text{dB}} - |H(\omega_0)|_{\text{dB}} = -3\text{dB}$$

-
- **Transfer Function and Frequency Response**
 - **Filters**
 - **Series/Parallel Resonance**

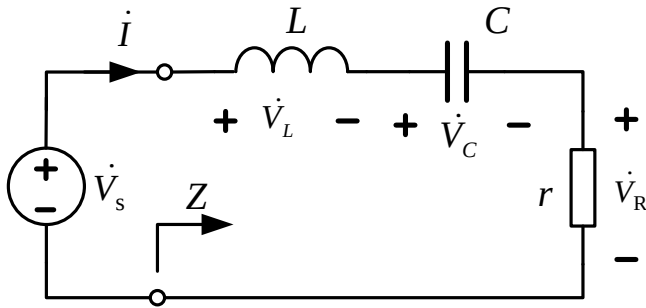
RLC in Series and in Parallel



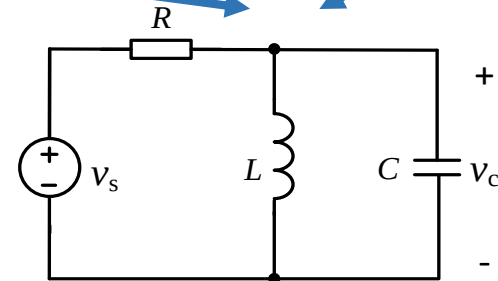
(a) **Lowpass** (b)



(a) **Highpass** (b)

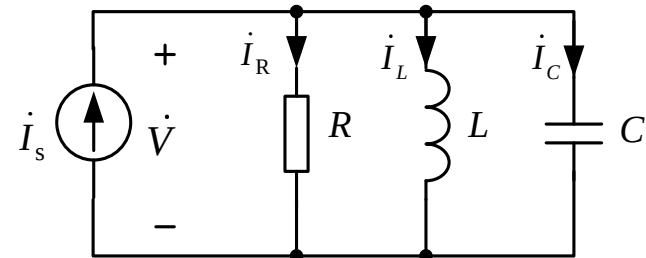


Bandpass



Bandpass

Natural Frequency: $\omega_0 = \frac{1}{\sqrt{LC}}$



Frequency Response of RLC in Series

- Voltage Ratio Transfer**

$$H(j\omega) = k \frac{\frac{\omega_0}{Q} j\omega}{(j\omega)^2 + \frac{\omega_0}{Q} j\omega + \omega_0^2}$$

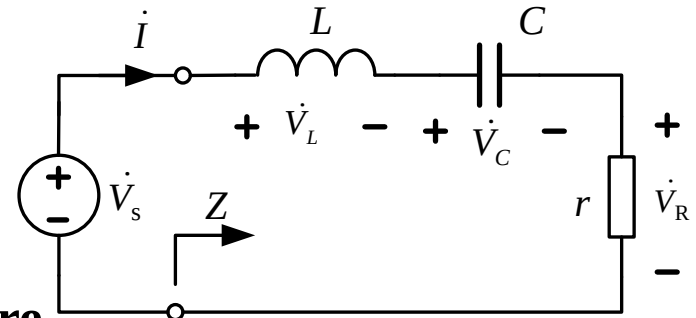
**Compare
Band-pass**

$$H(j\omega) = \frac{\dot{V}_R}{\dot{V}_s} = \frac{r}{r + j\omega L + \frac{1}{j\omega C}} = \frac{\frac{r}{L} j\omega}{(j\omega)^2 + \frac{r}{L} j\omega + \frac{1}{LC}}$$

ω_0^2

$$B = \omega_2 - \omega_1 = \frac{\omega_0}{Q} = \frac{r}{L} = \omega_0^2 Cr$$

$\frac{\omega_0}{Q}$

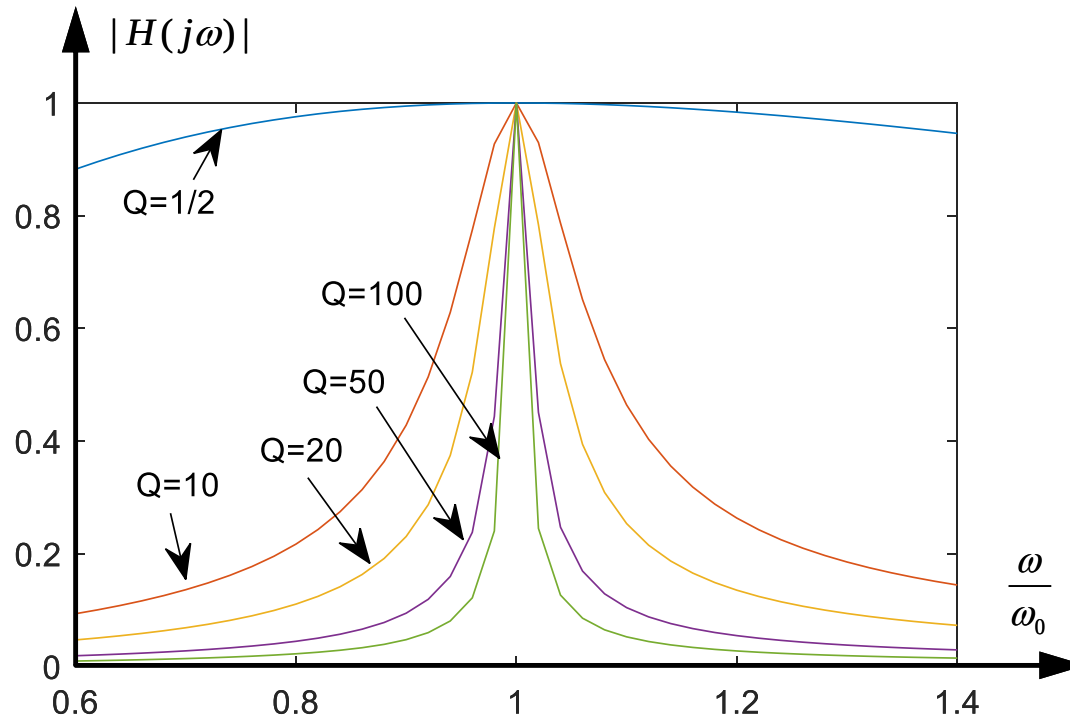


‘high Q ’ circuit: $Q > 10$, $B \ll \omega_0$, ‘narrow band’ circuit

$$\omega_1, \omega_2 \approx \omega_0 \mp \frac{B}{2}$$

The Normalization Amplitude Frequency Characteristics in RLC Series Circuits

- The influence of Q in circuits $B = \omega_2 - \omega_1 = \frac{\omega_0}{Q}$



The higher the value of Q , the more selective the circuit is but the smaller the bandwidth.

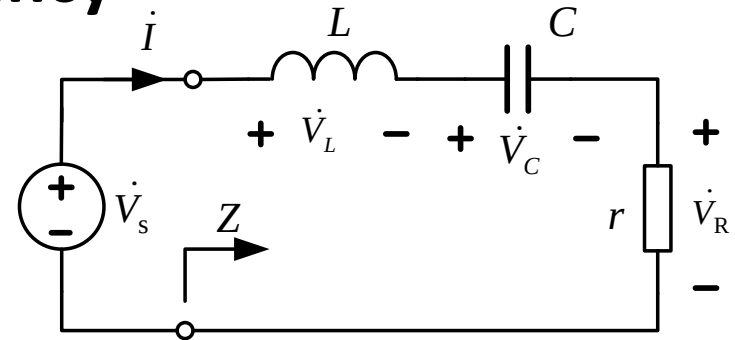
RLC Series Resonance

- **Resonance Condition and Frequency**

$$Z(\omega) = \frac{\dot{V}_s}{\dot{I}} = r + j(\omega L - \frac{1}{\omega C})$$

Condition: $\text{Im}[Z] = 0$

Resonance Frequency: $\omega_0 = \frac{1}{\sqrt{LC}}$



- **Resonance characteristics**

1. The impedance is purely resistive, thus, $Z = R$.

The reactance is called characteristic impedance:

$$\rho = \omega_0 L = \frac{1}{\omega_0 C} = \sqrt{\frac{L}{C}}$$

2. The voltage V_s and the current I are in phase, $\lambda_{pf} = \cos\theta = 1$.

3. $|Z(\omega)| = r$ is minimum.

4. V_L and V_C can be much more than V_s .

Series Resonance Quality Factor

- **Definition of Q**

$$Q = 2\pi \frac{\text{Peak energy stored in the circuit}}{\text{Energy dissipated by the circuit in one period at resonance}}$$

- **Q in RLC series resonance circuits**

$$Q = 2\pi \frac{LI_m^2 / 2}{TrI_m^2 / 2} = \frac{\omega_0 L}{r} = \frac{1}{\omega_0 Cr} = \frac{\rho}{r}$$

- **V_C and V_L in RLC series resonance circuits**

$$V_C = V_L = \frac{V_S}{r} \rho = QV_S \quad \text{'high } Q' : Q > 10$$

- **Q and B in RLC series resonance circuits**

$$B = \omega_2 - \omega_1 = \frac{\omega_0}{Q} = \frac{r}{L} = \omega_0^2 Cr$$

Example In RLC series resonance circuit, $L=1\text{H}$, $C=1\mu\text{F}$, find resonance frequency and characteristic impedance. Calculate quality factor, bandwidth and cutoff frequency when $r=10\ \Omega$ and $r=100\Omega$.

solution $\omega_0 = 1/\sqrt{LC} = 10^3 \text{ rad/s}$ $\rho = \sqrt{\frac{L}{C}} = \sqrt{\frac{1}{10^{-6}}} = 1 \text{ k}\Omega$

(1) $r=10\ \Omega$

$$Q = \frac{\rho}{r} = \frac{10^3}{10} = 100 \quad B = \frac{\omega_0}{Q} = \frac{10^3}{100} = 10 \text{ rad/s}$$

$$\omega_1 \approx \omega_0 - \frac{B}{2} = 995 \text{ rad/s} \quad \omega_2 \approx \omega_0 + \frac{B}{2} = 1005 \text{ rad/s}$$

(2) $r=100\ \Omega$

$$Q = \frac{\rho}{r} = \frac{10^3}{100} = 10 \quad B = \frac{\omega_0}{Q} = \frac{10^3}{10} = 100$$

$$\omega_1 \approx \omega_0 - \frac{B}{2} = 950 \text{ rad/s} \quad \omega_2 \approx \omega_0 + \frac{B}{2} = 1050 \text{ rad/s}$$

Quality Factor for a real inductor

- **Definition**

$$Q = 2\pi \frac{\text{Peak energy stored in the element}}{\text{Energy dissipated by the element in one period}}$$

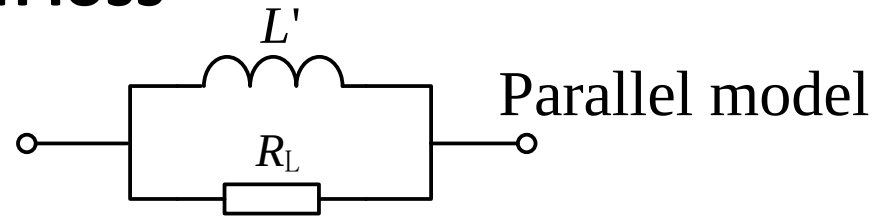
- **Two models for inductor with loss**

Series model



(a)

$$Q_L = 2\pi \frac{\frac{1}{2} L I_m^2}{T \cdot \frac{1}{2} r_L I_m^2} = \frac{\omega L}{r_L}$$



Parallel model

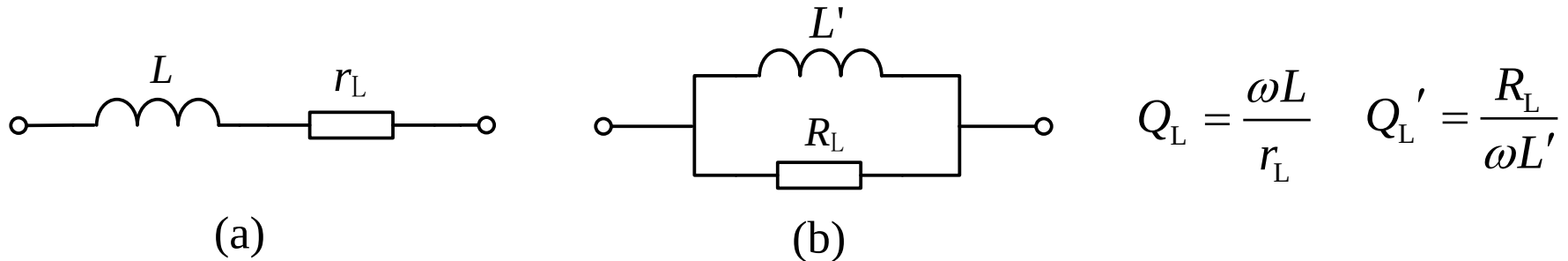
(b)

$$Q_L' = 2\pi \frac{\frac{1}{2} L' \left(\frac{U_m}{\omega L'} \right)^2}{T \frac{1}{2} \frac{U_m^2}{R_L}} = \frac{R_L}{\omega L'}$$

- (1) Resistance can be calculated after testing L and Q;
- (2) Q is related to frequency but keeps constant in a certain range.

Quality Factor for a real inductor

- Series model and parallel model have the same Q value when Q is very high



Equivalence for the two models:

$$Y_{\text{series}} = \frac{1}{r_L + j\omega L} = \frac{r_L}{r_L^2 + (\omega L)^2} - j \frac{\omega L}{r_L^2 + (\omega L)^2} \quad Y_{\text{parallel}} = \frac{1}{R_L} - j \frac{1}{\omega L'}$$

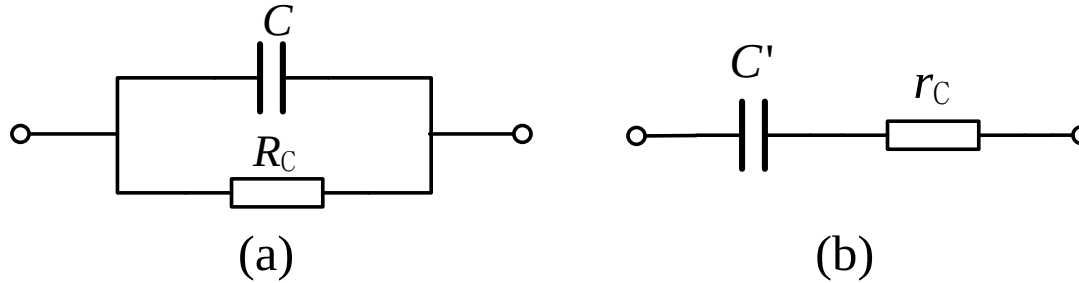
$$R_L = \frac{r_L^2 + (\omega L)^2}{r_L} = (1 + Q_L^2)r_L \approx Q_L^2 r_L = \frac{(\omega L)^2}{r_L}$$

$$\omega L' = \frac{r_L^2 + (\omega L)^2}{\omega L} = \left(\frac{1}{Q_L^2} + 1 \right) \omega L \approx \omega L$$

When Q is high

Quality Factor for a real capacitor

- Two models for capacitor with loss



Parallel model:

$$Q_C = 2\pi \frac{\frac{1}{2} C V_m^2}{T \cdot \frac{1}{2} V_m^2 / R_C} = \omega C R_C$$

When Q is high

$$C' = C \quad r_C = \frac{1}{\omega C Q_C}$$

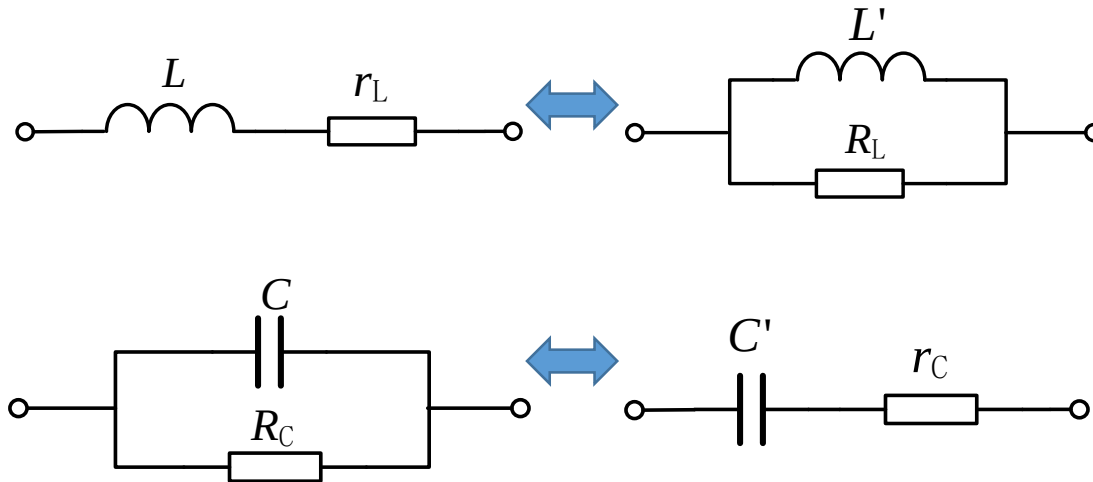
$$R_C = \frac{1}{(\omega C)^2 r_C} = Q_C^2 r_C$$

Summary for Quality Factor of a real reactor

- Series resistance r , parallel resistance R , reactance X

Under high Q , $Q = Q'$, $L = L'$, $C = C'$, and

$$r = \frac{|X|}{Q} \qquad R = Q|X| \qquad R = \frac{X^2}{r} = Q^2 r$$

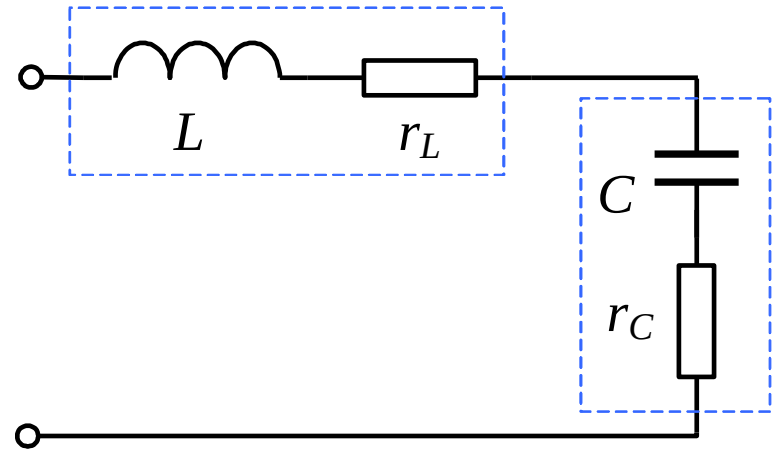


Q Value in Series Resonance

Considering only the loss in LC circuit

$$r_L = \frac{X_L}{Q_L} = \frac{\rho}{Q_L}$$

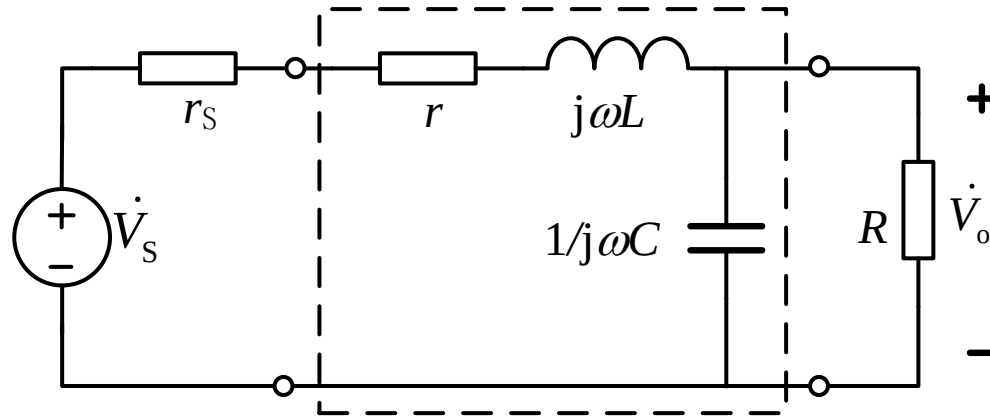
$$r_C = \frac{|X_C|}{Q_C} = \frac{\rho}{Q_C}$$



Q for the circuit:
$$Q = \frac{\rho}{r} = \frac{\rho}{r_L + r_C} = \frac{\rho}{\frac{\rho}{Q_L} + \frac{\rho}{Q_C}} = \frac{Q_L Q_C}{Q_L + Q_C}$$

Normally, $Q_C \gg Q_L$ **so** $Q \approx Q_L$

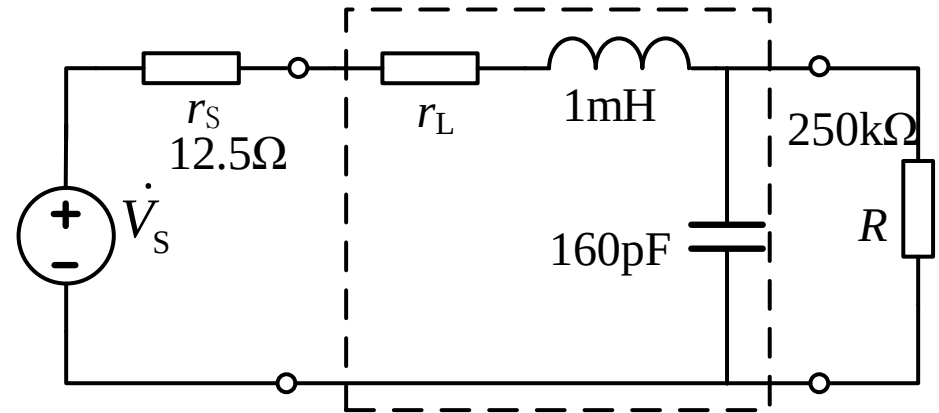
Effect on Q From Source and Load



- **Internal resistance in source and load increases the loss in circuit;**
- **Analysis method: take all resistors into equivalent series resistance in LC circuit.**

Example Given $Q_L=200$, $Q_C \gg Q_L$, $V_s=10\text{mV(rms)}$, the circuit works in resonance. Find Q and V_{C0} with or without load.

solution



(1) no-load , $R \rightarrow \infty$ and $r_s = 0$

$$Q \approx Q_L = 200 \quad V_{C0} = QV_s = 200 \times 10 \times 10^{-3} = 2 \text{ V}$$

(2) on load and r_s

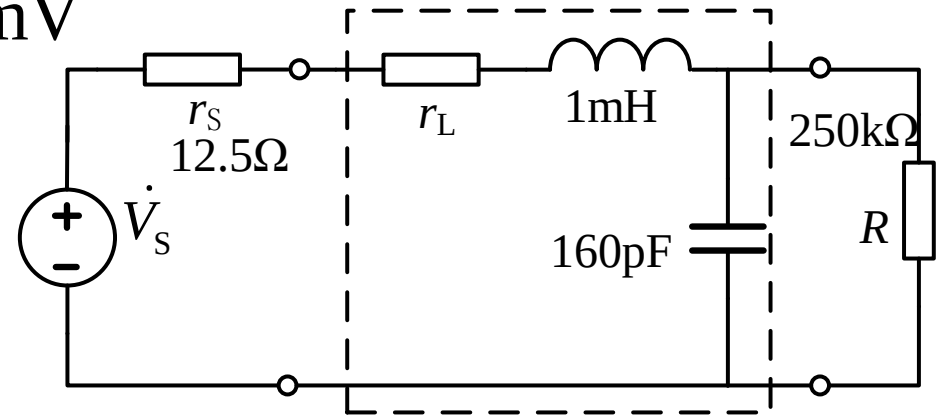
$$r_L = \frac{\rho}{Q_L} = \frac{\sqrt{L/C}}{Q_L} = \frac{\sqrt{10^{-3} / 160 \times 10^{-12}}}{200} = \frac{2.5 \times 10^3}{200} = 12.5 \Omega$$

$$Q_L=200, \quad Q_C \gg Q_L, \quad V_s=10\text{mV}$$

(2) on load $r_L = 12.5 \, \Omega$

Applying parallel model for Q_C :

$$Q_C = \omega C R_C = \frac{R_C}{|X_C|} = \frac{250\text{k}\Omega}{2.5\text{k}\Omega} = 100 \rightarrow \text{high } Q$$



Convert parallel model to series model:

$$r_C = \frac{X_C^2}{R} = \frac{\rho^2}{R} = \frac{(2.5 \times 10^3)^2}{250 \times 10^3} = 25 \, \Omega$$

Quality factor in on-load circuit:

$$Q' = \frac{\rho}{r_s + r_L + r_C} = \frac{2.5 \times 10^3}{12.5 + 12.5 + 25} = 50$$

$$V_{C0} = Q' V_s = 0.5 \, \text{V}$$

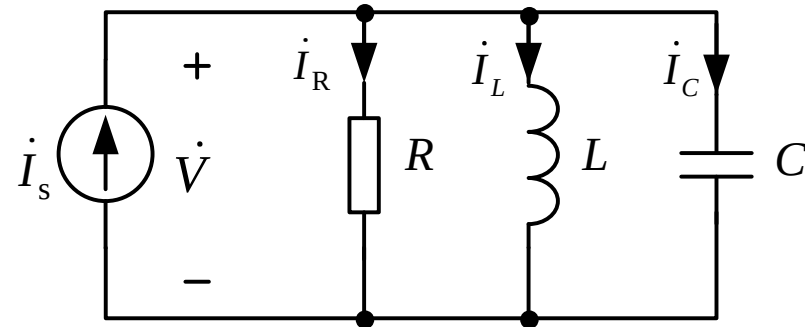
RLC Parallel Resonance

- **Resonance Condition and Frequency**

$$Y_{\text{in}}(\omega) = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$$

Condition: $\text{Im}[Y] = 0$

Resonance Frequency: $\omega_0 = \frac{1}{\sqrt{LC}}$



- **Resonance characteristics**

1. The impedance is purely resistive, thus, $Z = R$.

The characteristic impedance: $\rho = \omega_0 L = \frac{1}{\omega_0 C} = \sqrt{\frac{L}{C}}$

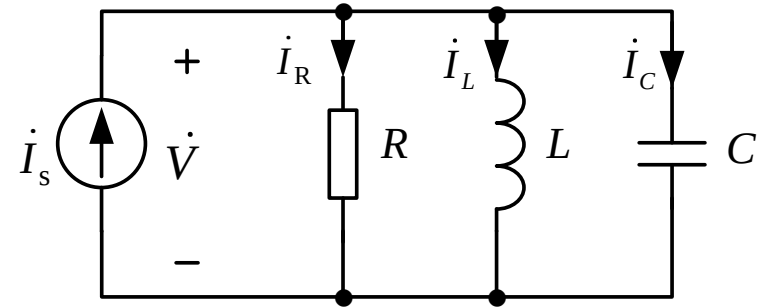
2. The voltage V and the current I_s are in phase, $\lambda_{\text{pf}} = \cos\theta = 1$.

3. $|Y(\omega)| = 1/R$ is minimum, $V \rightarrow \text{max}$.

4. I_L and I_C can be much more than I_s .

RLC Parallel Resonance

- **Quality factor in parallel resonance:**



$$Q = 2\pi \frac{\frac{1}{2} C V_m^2}{T \cdot \frac{1}{2} V_m^2 / R} = \omega_0 C R = \frac{R}{\omega_0 L} = \frac{R}{\rho}$$

- **Current in parallel resonance :**

$$I_C = I_L = \frac{I_s R}{\omega_0 L} = I_s \omega_0 C R = Q I_s$$

$$I_R = I_s$$

RLC Parallel Resonance

- Input impedance:

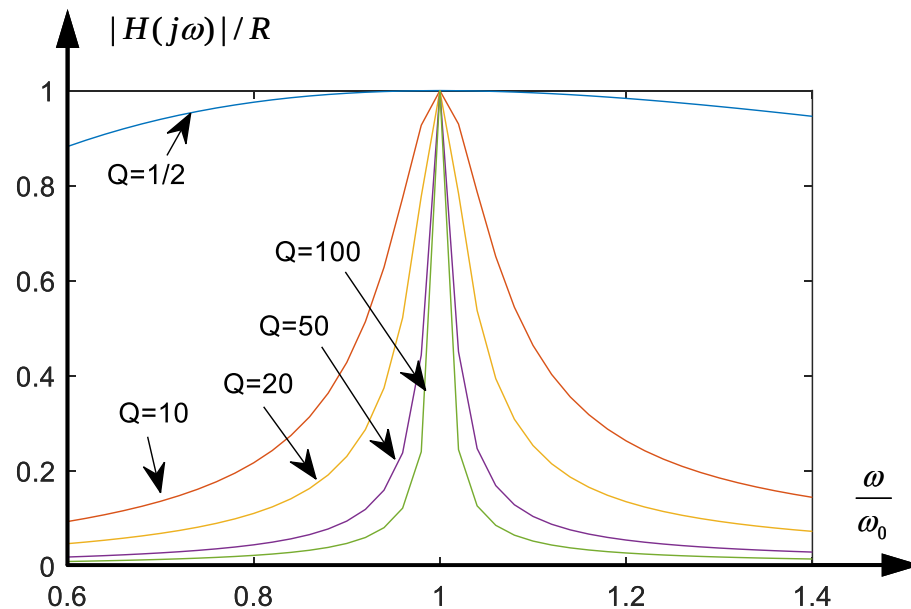
$$H(j\omega) = \frac{\dot{V}}{\dot{I}_s} = Z(j\omega) = \frac{1}{\frac{1}{R} + j\omega C + \frac{1}{j\omega L}} = \frac{\frac{1}{C} j\omega}{(j\omega)^2 + \frac{1}{RC} j\omega + \frac{1}{LC}}$$

- Bandpass characteristics:

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad Q = R\omega_0 C = R/(\omega_0 L) = R/\rho = \omega_0/B \quad k=R$$

$$\frac{\omega_0}{Q}$$

- Frequency selection and Q :



Practical Parallel Resonance

$$Y = j\omega C + \frac{1}{r + j\omega L} = j\omega C + \frac{r - j\omega L}{r^2 + (\omega L)^2}$$

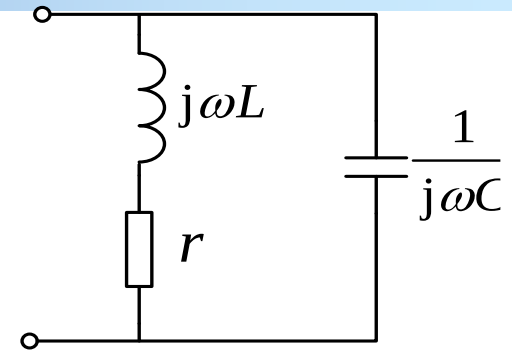
Set $\text{Im}[Y] = \omega C - \frac{\omega L}{r^2 + (\omega L)^2} = 0$

Get parallel resonance frequency:

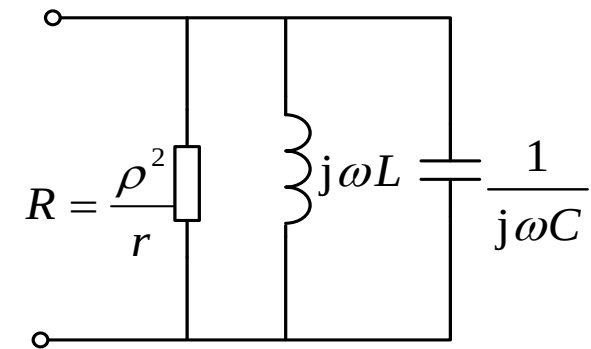
$$\omega_p = \frac{1}{\sqrt{LC}} \sqrt{1 - \frac{r^2}{\rho^2}} = \omega_0 \sqrt{1 - \frac{1}{Q_L^2}}$$

When Q is high, $\omega_p \approx \omega_0$

$$Q = \frac{R}{\rho} = \frac{\rho}{r} = \frac{\sqrt{L/C}}{r}$$

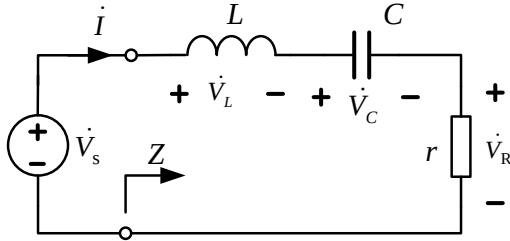


$$R = \frac{X^2}{r} = \frac{\rho^2}{r} = \frac{L}{Cr}$$



For high Q inductor, $r \ll \rho$

Recap series and parallel resonance



$$H(j\omega) = \frac{\dot{V}_R}{\dot{V}_s} = \frac{\frac{r}{L} j\omega}{(j\omega)^2 + \frac{r}{L} j\omega + \frac{1}{LC}}$$

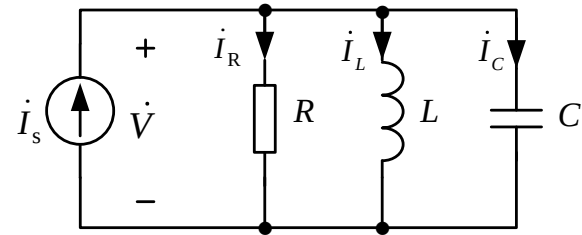
$$\omega_0 = \frac{1}{\sqrt{LC}} \quad B = \frac{r}{L}$$

$$\rho = \omega_0 L = \frac{1}{\omega_0 C} = \sqrt{\frac{L}{C}}$$

$$Q = \frac{\rho}{r} = \frac{\sqrt{L/C}}{r} = \frac{\omega_0}{B}$$

$$Q > 10, \quad B \ll \omega_0, \quad \omega_1, \omega_2 \approx \omega_0 \mp \frac{B}{2}$$

$$V_C = V_L = \frac{V_s}{r} \rho = QV_s$$



$$H(j\omega) = \frac{\dot{V}}{\dot{I}_s} = \frac{\frac{1}{C} j\omega}{(j\omega)^2 + \frac{1}{RC} j\omega + \frac{1}{LC}}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad B = \frac{1}{RC}$$

$$\rho = \omega_0 L = \frac{1}{\omega_0 C} = \sqrt{\frac{L}{C}}$$

$$Q = \frac{R}{\rho} = \frac{R}{\sqrt{L/C}} = \frac{\omega_0}{B}$$

$$\omega_1, \omega_2 \approx \omega_0 \mp \frac{B}{2}$$

$$I_C = I_L = \frac{I_s R}{\rho} = QI_s$$

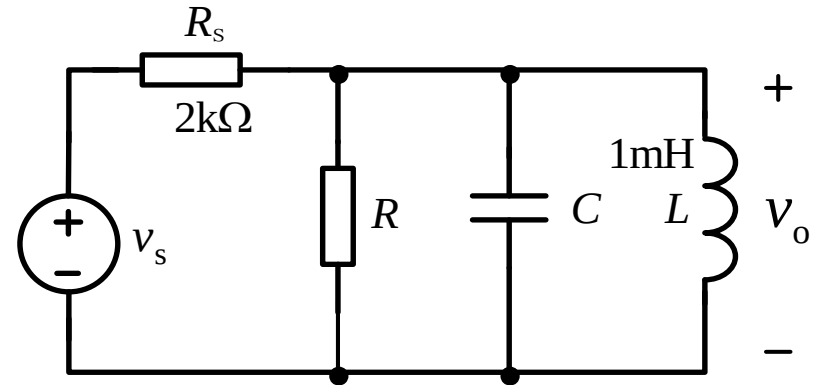
Example Given $\omega_0 = 10^5 \text{ rad/s}$. If required bandwidth is not more than $B = 10^4 \text{ rad/s}$, find C and minimum R .

solution From: $\omega_0 = \frac{1}{\sqrt{LC}}$

Get $C = \frac{1}{\omega_0^2 L} = 100 \text{ nF}$

Total parallel resistance in the circuit is

$$R_s // R = \frac{R_s \cdot R}{R_s + R}$$



Since bandwidth is not more than $B = 10^4 \text{ rad/s}$, Q is not less than

$$Q = \frac{\omega_0}{B} = 10 \quad \rightarrow \quad R_s // R = Q\omega_0 L = 1000$$

$$R \geq 2 \text{ k}\Omega$$

Example Find the resonant frequency, no-load Q , on load Q and B.

solution

No-load Q:

$$Q = \frac{\sqrt{L/C}}{r} = \frac{\sqrt{0.244/0.5 \times 10^{-6}}}{32} = 21.8$$

$Q > 10$, high Q

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{6.28\sqrt{0.244 \times 0.5 \times 10^{-6}}} = 456 \text{ Hz}$$

Convert resistor in series with inductor to that in parallel:

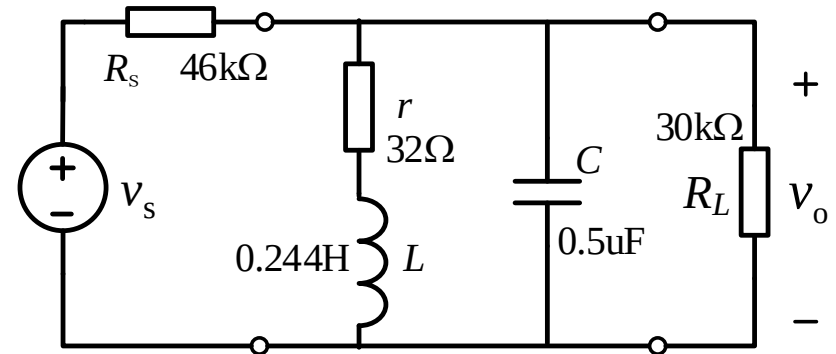
$$R = \frac{\rho^2}{r} = \frac{L}{Cr} = \frac{0.244}{0.5 \times 10^{-6} \times 32} = 15.3 \text{ k}\Omega$$

Bandwidth :

$$B_f = \frac{f_0}{Q'} = \frac{456}{11.8} = 38.6 \text{ Hz}$$

On load Q:

$$Q' = \frac{R_{//}}{\rho} = \frac{1}{\frac{1}{R} + \frac{1}{R_s} + \frac{1}{R_L}} \frac{1}{\sqrt{L/C}} = \frac{1}{\left(\frac{1}{15.3\text{k}} + \frac{1}{46\text{k}} + \frac{1}{30\text{k}}\right)699} = 11.8$$



Homework

- **15, 17, 19, 20, 21**